

P1-Basic structure manipulation with BioPython

ex_1.py

input: python3 ex_1.py 1ubq 3.8

- required arguments:
 - **PDB_file_name:** 1ubq
 - **distance** : 3.8 (measured in armstrongs)

Terminal output: Structure exists: '.\pdb1ubq.ent'

Performing Neighbour search, with max distance set to: 3.8 ...

{('PRO', 'SER'): (144, 151, 3.8),

...

('THR', 'GLY'): (69, 76, 3.77)}

Output → pair of residues whose carbon atoms are closer than the set distance:

{ (residue1 , residue2) :

(CA_{r1} serial number, CA_{r2} serial number, distance between CA atoms - Å) }

ex_2.py

input: python3 ex_2.py 1ubq 2

- required arguments:
 - **PDB_file_name:** 1ubq
 - **residue number:** 2 (measured in armstrongs)
 - **--chain_number:** optional argument

Terminal output: Structure exists: '.\pdb1ubq.ent'

Residue 2 from 1ubq has these atoms:

[<Atom N>, <Atom CA>, <Atom C>, <Atom O>, <Atom CB>, <Atom CG>, <Atom CD>,
<Atom OE1>, <Atom NE2>]

Output → List of atoms of the set residue on the PDB structure.

ex3.py

input: ex_3.py 1ubq

- required arguments:
 - **PDB_file_name:** 1ubq
 - **--cut_off_distance:** (optional) None (default set to 3.5)

output: Structure exists: '.\pdb1ubq.ent'

{ ('HOH113: O', 'HOH130: O'): 2.6,

...

('LEU8: O', 'THR9: N'): 2.28}

Output → pair of polar atoms that form a hydrogen bond:

{ (atom1 , atom2) : distance between atoms - Å }')

ex_4.py

input: ex_4.py 1ubq GLY

- required arguments:
 - **PDB_file_name:** 1ubq
 - **residue_type:** GLY

output: Structure exists: '.\pdb1ubq.ent'

{'GLY 10, CA': array([28.978, 43.96 , 14.678], dtype=float32),

...

'GLY 76, CA': array([40.373, 39.813, 33.944], dtype=float32)}

Output -> selected residue and its carbon atoms location:

{ (selected_residue , CA) : [CA atom coordinate array] }

ex_5.py

input: ex_5.py 1ubq

- required arguments
 - **PDB_file_name:** 1ubq
 - **--cut_off_distance:** (optional) None (default set as 2.5)

output: Structure exists: '.\pdb1ubq.ent'

{('Carbon residue MET1: C', 'Nitrogen residue MET1: N'): 2.43,

...

('Carbon residue GLY76: C', 'Nitrogen residue GLY76: N'): 2.44 }

Output -> pairs of C and N atoms from different residues closer than the cut-off, list of backbone connectivity, residues linked by peptide bonds.

{ (C atom, N atom) : distance in Å }

ex_6.py

input: ex_6.py 2mbe

- required arguments
 - **PDB_file_name:** 2mbe
 - **--cut_off_distance:** (optional) None (default set as 2.05)

output: As cut-off distance was set as None, default value will be used: 2.05

Structure exists: '.\pdb2mbe.ent'

{('CYS2: SG', 'CYS2: SG'): 0.79}

Output -> pair of sulfur atoms that form a disulfide bond:

{ (residue1 id1 : sulfur atom forming the disulfide bond, residue2 id2 : sulfur atom) : distance of the bond - Å }

ex_7.py

input: ex_7.py 2mbe GLY TRP

- required arguments:
 - **PDB_file_name:** 2mbe
 - **residue1** : GLY
 - **residue2** : TRP

Output: Structure exists: '.\pdb2mbe.ent'

{('GLY46: N', 'TRP10: N'): 20.19,

...

('GLY60: HA3', 'TRP10: HH2'): 32.74}

Output -> distance between all of the atoms of 2 given residues

{ (residue1+id: atom+id , residue2+id:atom+id) : distance between atoms - Å }